EARTHPRESS · USER REQUIREMENTS

EarthPress user requirements

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First user requirements (kit 1.8 pipeline), drawn from the users and stakeholders in EPR-PRB-001 Tables 1 and 2 and the existing requirements EPR-REQ-001 v0.8

EarthPress user requirements

What users need, written from their side. Product requirements that meet these needs are in EPR-PRS-001 (`03-requirements.md`) and the links are in the traceability matrix EPR-RTM-001 (`03b-traceability.md`). They are drawn from desk research and the earlier requirements; none has yet been confirmed with users, which is the first task with the co-design partner (EPR-PRB-001). Method: STANDARDS section 22.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Standard block size and height	Every block is the same size and height, so walls run true and mortar joints stay thin	High	Masons and builders (PRB Table 1)
UR-002	Performance	Dense, strong blocks	Blocks are pressed hard enough to be strong and durable when the soil and mix are right	High	Self-builders; NGO trainers; architects (PRB Table 1)
UR-003	Performance	Output for a house	A small crew makes enough blocks in a day to build a small house in about a week of pressing	High	Block-making enterprises; self-builders (PRB Table 1)
UR-004	Usability	Ordinary adults can operate it	Ordinary adults, including first-time volunteers, can press blocks without straining, two people sharing the work if needed	High	Self-builders; NGO trainers (PRB Table 1)
UR-005	Deployment and assembly	Built and repaired by a local welder	A local welder can build and repair the press from common steel with a stick welder, grinder and drill press	High	Local welders and fabricators (PRB Table 1)
UR-006	Logistics	Movable between sites	The press can be moved between sites on a pickup and carried in pieces by two people	Medium	NGO trainers; self-builders (PRB Table 1)
UR-007	Maintenance	Long life, replaceable wear parts	The press lasts many thousands of blocks, and the parts that wear can be replaced with hand tools	Medium	Block-making enterprises (PRB Table 1)
UR-008	Usability	Mix chosen with field tests	Builders can tell whether a soil is suitable and how much stabilizer it needs, with simple field tests and no laboratory	High	Self-builders; NGO trainers (PRB Table 1)
UR-009	Cost	Affordable press and kit	The press and soil kit cost a few hundred dollars in parts, far below hydraulic machines	High	Self-builders; NGOs; funders (PRB Tables 1 and 2)
UR-010	Environment and durability	Low stabilizer use	Blocks need little cement or lime, which is the main running cost	Medium	Self-builders; block-making enterprises (PRB Table 1)
UR-011	Environment and durability	Lower carbon than fired brick	Walls made with the blocks carry much less embodied carbon than fired brick	Medium	Architects; funders (PRB Tables 1 and 2)
UR-012	Safety	Safe for operators and bystanders	Nobody is hurt by the lever, the lid, pinch points, cement or dust when the press is used as instructed	High	NGO trainers; funders; all users (PRB Tables 1 and 2)
UR-013	Regulatory	Open documentation	Drawings, parts list and build plan are open and complete enough to build without contacting the author	Medium	Local welders; NGOs (PRB Table 1)

ID	Category	Title	Need	Priority	Source
UR-014	Usability	Human powered	The press needs no fuel, electricity or hydraulics on site	High	Self-builders on rural sites (PRB Table 1)
UR-015	Environment and durability	Works outdoors in dust and rain	The press keeps working outdoors in dust, heat and rain	Medium	Self-builders; block-making enterprises (PRB Table 1)
UR-016	Regulatory	Blocks testable against a strength reference	Blocks can be tested against a recognized strength reference for the intended use	Medium	Architects; building authorities (PRB Tables 1 and 2)

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.